

$$\text{Freq. of } C0 = 16.35 \text{ Hz}$$

dist + 12 · octave

$$\text{Freq.} = 16.35 \cdot \left(12\sqrt[2]{2}\right)$$

$$\frac{\text{freq.}}{16.35} = \left(12\sqrt[2]{2}\right)^{\text{dist} + 12 \cdot \text{octave}}$$

dist = number assigned
with note
letter

$$\log_2 \left(\frac{\text{freq.}}{16.35} \right) = \frac{\text{dist} + 12 \cdot \text{octave}}{12}$$

$$12\sqrt[2]{2} \approx 19.55$$

octave = octave num

$$12 \log_2(\text{freq.}) - 12 \log_2(16.35) = \text{dist} + 12 \cdot \text{octave}$$

$$\text{dist} = 12 \left(\log_2(\text{freq.}) - \log_2(16.35) - \text{octave} \right)$$

Octave:

$$\frac{\text{freq.}}{16.35} = (12\sqrt{2})^{\text{dist} + 12 \cdot \text{octave}}$$

$$\log_2(\text{freq.}) - \log_2(16.35) = \frac{\text{dist} + 12 \cdot \text{octave}}{12}$$

$$12 \log_2(\text{freq.}) - 12 \log_2(16.35) = \text{dist} + 12 \cdot \text{octave}$$

$$\text{octave} = \log_2(\text{freq.}) - \log_2(16.35) - \frac{\text{dist}}{12}$$

As octave must be an integer,
and $\frac{\text{dist}}{12}$ is not significant within
the musical range of notes,
octave can be approximated with:

$$\text{octave} = \text{trunc}(\log_2(\text{freq.}) - 4)$$